

Lab Record: N-Queen's Problem Using Backtracking

Aim

To design and implement a C program for solving the N-Queen's problem using Backtracking.

Theory

The N-Queen's problem is a classic problem in backtracking where N queens must be placed on an $N \times N$ chessboard such that no two queens attack each other. A queen can attack horizontally, vertically, and diagonally. Backtracking is used to place queens row by row and backtrack whenever a conflict occurs.

Algorithm

1. Start with an empty chessboard of size $N \times N$.
2. Place queens row by row beginning from the first row.
3. For each row, check every column:
 - a. Check whether the queen can be placed safely.
 - b. A position is safe if:
 - i. No queen exists in the same column.
 - ii. No queen exists in the left diagonal.
 - iii. No queen exists in the right diagonal.
4. If the position is safe:
 - a. Place the queen at that position.
 - b. Recursively move to the next row.
5. If all queens are placed successfully, display the solution.
6. If no valid position is found in a row:
 - a. Remove the previously placed queen (Backtracking).
 - b. Try the next possible position.
7. Repeat the process until all possible solutions are checked.
8. Stop when all solutions are generated or the required solution is found.

Time Complexity

Worst Case: $O(N!)$

Space Complexity: $O(N^2)$

Applications

- Artificial Intelligence
- Constraint Satisfaction Problems
- Game Development
- Puzzle Solving
- Optimization Problems

Conclusion

The N-Queen's problem demonstrates the use of Backtracking technique to solve constraint-based problems efficiently by exploring and undoing choices.